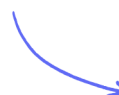


Structured Questions

Electrons & Ions

Endothermic Ionisation Energy / Electronic Structures / Shapes of Orbitals /
Electronic Configurations & Chemical Properties

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Total Marks

/14

1 (a) This question is about copper and its compounds.

Complete the electronic configurations of Cu and Cu²⁺.

Cu [Ar].....

Cu²⁺ [Ar].....

(2 marks)

(b) A sample of copper contains the isotopes ⁶³Cu and ⁶⁵Cu.

i) Complete the table to show the numbers of subatomic particles in the atoms of these two isotopes of copper.

(2)

Isotope	Protons	Neutrons	Electrons
⁶³ Cu			
⁶⁵ Cu			

ii) Explain the term isotopes, using the information in the table.

(2)

iii) State why the two isotopes of copper have the same chemical reactions.

(1)

iv) The relative atomic mass of copper in this sample is 63.4. Calculate the percentage abundances of the isotopes ⁶³Cu and ⁶⁵Cu in this sample.

You **must** show your working.

(2)

(7 marks)

(c) Copper(II) sulfate, CuSO_4 , can be made by reacting solid copper(II) carbonate with dilute sulfuric acid.

i) Write an equation for the reaction that occurs. State symbols are not required.

(1)

ii) An experiment was carried out to produce pure, dry crystals of hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Copper(II) carbonate was mixed with 50.0 cm^3 of 1.00 mol dm^{-3} sulfuric acid until no more reacted. The mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ obtained was 10.87 g .

Calculate the percentage yield for this reaction, giving your answer to an appropriate number of significant figures.

[Molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} = 249.6 \text{ g mol}^{-1}$]

(4)

(5 marks)